2025-07-21 - Analyze the data and perform interval testing for sampling distributions.

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⇒ Sample of 11 circuits from a large normal pop has a mean resistance of 2.20 Ω. We know from past testing that pop S.D is 0.35 Ω. Determine a 95% C.I for the mean resistance of the population.

Same Distributions

$$\Rightarrow \left(\bar{x} - z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \; , \; \bar{x} + z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}\right)$$

Level of Significance

Standard Error

$$= \left(2.20 - 1.96 \times \frac{0.35}{\sqrt{11}} \; , \; 2.20 + 1.96 \times \frac{0.35}{\sqrt{11}}\right)$$

$$(1.9932 \; , \; 2.4068)$$

⇒ σ is not known:

=> σ is not known

Pop S.D is replaced by unbiased estimates.

$$s_1 = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \bar{x})^2$$

$$\left(\bar{x} - z_{\alpha/2} \times \frac{s_1}{\sqrt{n}} , \; \bar{x} + z_{\alpha/2} \times \frac{s_1}{\sqrt{n}} \right)$$

=> Small samples $(n < 30)$

Student's T-Distribution

$$\left(\bar{x} - t_{\alpha/2} \times \frac{s_1}{\sqrt{n}} , \; \bar{x} + t_{\alpha/2} \times \frac{s_1}{\sqrt{n}} \right)$$

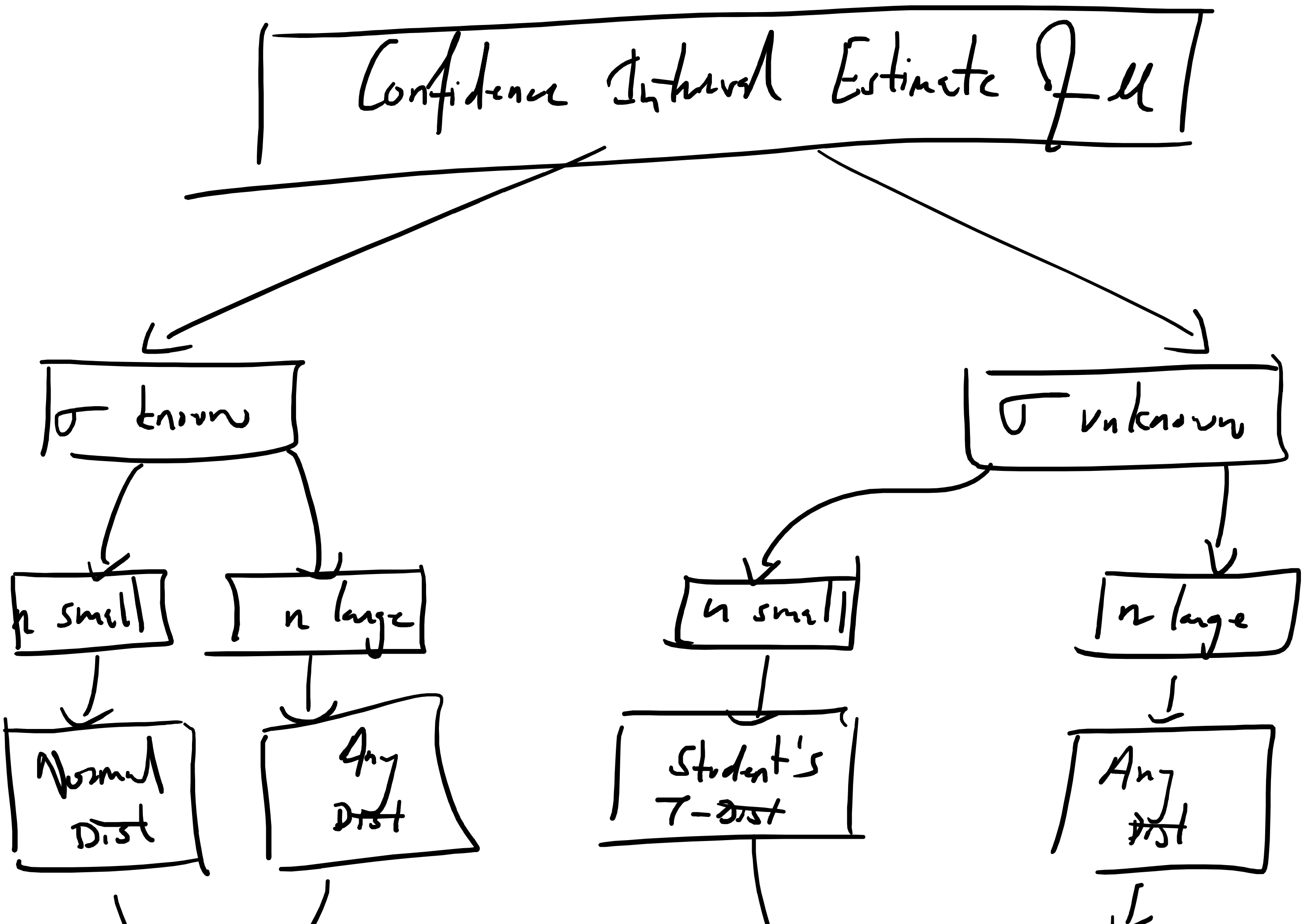
=> A sample of 11 circuits with mean resistance of 2.20 Ω. Pop S.D is not known. s_1 is 0.35 Ω. Determine CI at 95%.

$t_{\alpha/2} \approx 95\%$ CI $= \underline{2.228}$

$$\left(\bar{x} - t_{\alpha/2} \times \frac{s_1}{\sqrt{n}} \;,\; \bar{x} + t_{\alpha/2} \times \frac{s_1}{\sqrt{n}}\right)$$

$$\left(2.20 - 2.228 \times \frac{0.35}{\sqrt{11}} \;,\; 2.20 + 2.28 \times \frac{0.35}{\sqrt{11}}\right)$$

$$\underline{\left(1.964 \;,\; 2.335\right)}$$



$$\left(\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \;,\; \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}\right)$$

$$\left(\bar{x} - Z_{\alpha/2} \times \frac{s_1}{\sqrt{n}} \;,\; \ldots\right)$$

$$\left(\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}},\ \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}\right)$$

$$\left(\bar{x} - t_{\alpha/2} \times \frac{s_1}{\sqrt{n}},\ \bar{x} + Z_{\alpha/2} \times \frac{s_1}{\sqrt{n}}\right)$$

$$\left(\bar{x} - t_{\alpha/2} \times \frac{s_1}{\sqrt{n}},\ \bar{x} + t_{\alpha/2} \times \frac{s_1}{\sqrt{n}}\right)$$

⇒ **Estimating Population Variance (σ^2)**

⇒ $$s_1^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

⇒ $$s_2^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Note: For large sample $\left[\ s_1^2 = s_2^2\ \right]$

Case 1: Samples are small $(n < 30)$

Dist Stat $= \dfrac{(n-1)s_1^2}{\sigma^2}$ D-o-F $(n-1)$

⇒ **Interval Estimate**:

$$\left(\frac{(n-1)s_1^2}{\chi^2_{\alpha/2}},\ \frac{(n-1)s_1^2}{\chi^2_{1-\alpha/2}} \right)$$

$$\left(\chi^2_{\alpha/2} \quad , \quad \chi^2_{1-\alpha/2} \right)$$